

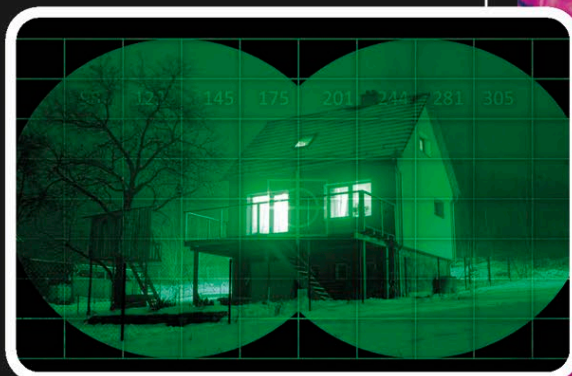
MEASURING HEAT

THERMAL IMAGING

All objects emit heat in the form of infrared radiation. This is invisible to the human eye, but thermal imaging cameras equipped with special sensors can detect it. These map the differences in temperatures and turn the readings into visible color images. Cool objects appear blue or purple, while warmer objects look red or yellow. These cameras are useful for monitoring people and detecting other objects when visibility is poor.

NIGHT VISION GOGGLES

Unlike thermal cameras, night vision goggles do not work in absolute darkness. They intensify any faint visible light and then translate it into a clearer green-tinted final image.



Some snakes can detect infrared radiation, allowing them to **hunt in the dark**.

MEASURING WAVES

Infrared is a form of electromagnetic radiation (see page 142), which, like the visible light we see, travels as waves. All waves are different, and we can measure them by looking at their wavelength (the distance between two peaks), their amplitude (the wave's height), and their frequency (the number of waves per second). Infrared waves cannot be seen by the naked eye, because they have a longer wavelength than visible light.

